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IEEE Chandigarh Subsection International Conference : Submission (1176) has been created.

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Microsoft CMT <noreply@msr-cmt.org> 29 April 2026 at 17:11
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Track Name: Artificial Intelligence and Data Science

Paper ID: 1176

Paper Title: Comparative Study of Machine Learning Regression Models for Student Performance Prediction

Abstract:
Since the 2000's a huge number of studies have emerged in the area of educational data mining that aims to help students by predicting the academic performance to flag students who may need help at school and university. This paper aims to compare and evaluate the performances of four supervised machine learning regression models, namely Linear Regression, Ridge Regression, Decision Tree Regressor and Random Forest Regressor, to find the best predictive model for student's grades. The UCI Student Performance Dataset (student-mat.csv) having a total of 395 student records further with 33 attributes representing demographic, social, and academic details with final mathematics grade G3 as target variable is used. The categorical variables were processed using LabelEncoder, data was divided into 80:20 with train-test-split and StandardScaler was used to normalize the data. The evaluation of all four models used MAE, MSE and R^2 score. The best model we found is the Random Forest Regressor with an R^2 of 0.8288 a MAE of 1.147 and a MSE of 2.838. Thus, we can conclude that this model can explain 83% of the variation in student final grades. Feature importance analysis revealed that G1 and G2, which are grades obtained in the first and second periods respectively, are the strongest predictors of G3. This validates the importance of interim records for building effective early warning systems.

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Secondary Subject Areas: Not Entered

Submission Files:
Research_Paper(Student_Performance_Prediction).pdf (351 Kb, Wed, 29 Apr 2026 11:23:28 GMT)

Submission Questions Response:
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